

TeleViewDemo: Experience the Future of 3D Teleconferencing

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ABSTRACT

Recent demonstrations of 3D telepresence provide a glimpse into the future where 2D video communication is replaced with photo-realistic virtual avatars rendered on 3D displays by using dedicated hardware. Our platform lets visitors experience real-time end-to-end 3D teleconferencing using commodity hardware. This demo integrates current state-of-the-art face reconstruction and rendering algorithms into an end-to-end system to illustrate the utility and capabilities of commodity telepresence on a range of 3D displays.

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1 INTRODUCTION

The COVID-19 pandemic has accelerated the shift to online meetings, leading to an increased demand for teleconferencing technology to offer a more realistic and in-person experience. However, for meeting visitors, feeling a sense of presence through in-person cues, such as, eye-contact and proximity awareness remains a major challenge. The availability of view-dependent displays, like HMDs, HUDs, and fish-tank virtual reality (FTVR) displays, used in VR/AR/Mixed reality, along with advances in neural rendering and 3D head reconstruction from single RGB cameras has set the stage for a new wave of algorithms to efficiently, with high frame rates and low-latency, encode and decoded 3D representations of meeting visitors. We have built an end-to-end 3D teleconferencing system, TeleViewDemo, that supports a variety of view-dependent display combinations to support testing their effects on 3D telepresence in real-world conditions. Our platform arises out of our own research to advance the state of the art of low-latency 3D head reconstruction and neural rendering. For this exhibit, visitors will get to experience the latest end-to-end 3D teleconferencing on commodity-hardware.

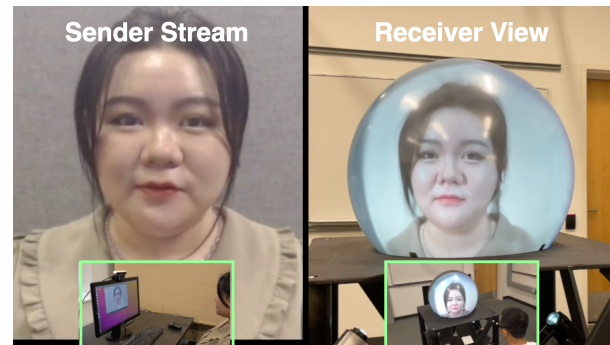


Figure 1: TeleViewDemo modular end-to-end 3D teleconferencing in action. The Sender side webcam image (left) is encoded to a neural 3D model. Its shape, expression, pose parameters are sent to the receiver side where a view-dependent rendering is shown on the Receiver's 3D display (right).

Contribution. Our TeleViewDemo system's major contribution is that it lets people experience 3D telepresence on different view-dependent displays, including flat, curved, and head-mounted screens. View-dependent displays require rendering a person's likeness from different views, hence, a 3D teleconferencing system needs to be able to capture and transmit the person's 3D characteristics. Some proprietary systems have been proposed to achieve this goal, such as Google's Starline project [Lawrence et al. 2021] and Microsoft's HoloPortation [Orts-Escolano et al. 2016], but each has either a closed system or large-scale proprietary or prohibitively expensive hardware. Further, as these are closed, researchers are not able to test downstream tasks, such as perceptual studies on the difference of the available 3D displays. We address this major obstacle and will bring a demonstration for visitors to Siggraph Asia 2022.

2 RELATED WORK

Research in teleconferencing has moved from 2D video to 3D. While significant research has gone into developing algorithms to make these systems feasible, we focus on the systems as a whole. Recent 3D teleconferencing systems use proprietary components. Pioneering systems have appeared over 20 years to achieve 3D teleconferencing, such as: a room-scale system by Gibbs et al. [1999], a convex fast-spinning mirror based display system by Jones et al.

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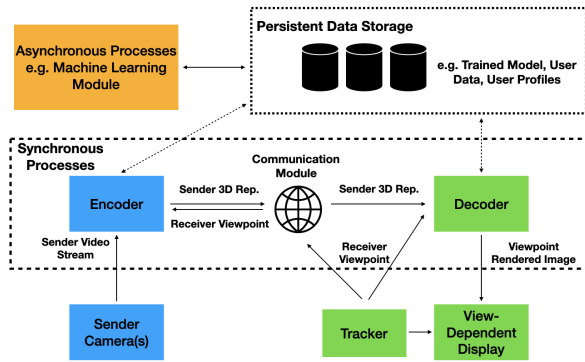


Figure 2: The main components needed by our 3D teleconferencing system are: 1. Sender/Receiver hardware, 2. Encoder, 3. Decoder, and 4. Persistent Data Storage needs.

[2009], a GPU-accelerated multi-Kinect based system with a parallax auto-stereoscope display [Maimone et al. 2012], a room-size multiple depth camera based system Dou et al. [2012] that includes eye-contact, and a system using an array of IR cameras and lasers with RGB and Microsoft Kinect depth cameras for three person teleconferencing by Zhang et al. [2013]. More recently, [Lawrence et al. 2021] provides an end-to-end system which utilizes an array of IR, RGB, and tracking cameras with an autostereoscopic display among other contributions to enable face-to-face teleconferencing. These ongoing research efforts motivate our work to provide a demonstration of these components playing together to form an end-to-end teleconferencing system. All the components in the TeleViewDemo are easily obtained allowing researchers to replicate the environment to test and compare performance of their own work in real-world 3D teleconferencing scenarios.

3 METHOD

We describe the scenario of one person (Sender) talking to another (Receiver) as shown in Figure 1. The roles are switched for bidirectional conversation.

Hardware Setup. The *Sender* side uses a single RGB camera (Logitech C920 Webcam HD Pro, 30 FPS) and one computer equipped with a GPU. The *Receiver* side uses: i) a view-dependent display such as: an FTVR display [Zhou et al. 2017], a VR headset, or a flatscreen FTVR display, ii) a computer equipped with a GPU, and iii) a head/eye tracking system. For this demonstration we will use a Polhemus Fastrak for the spherical FTVR display, a 3D depth camera (Intel RealSense D435) for the flat screen FTVR display and the IMU in the VR headset to illustrate a range of tracking technology can be integrated.

Encoder-Decoder. We use the DECA [Feng et al. 2021] *Encoder* and a Deferred Neural Rendering [Thies et al. 2019] *Decoder* for the encoding of the input video streams and the subsequent decoding and view dependent rendering. The *Encoder* generates a small set of 3D head parameters of the *Sender*. We use the FLAME [Li et al. 2017] parameters for our representation that includes: shape, expression, pose and color. The neural renderer continuously renders new views

given the current view positions of the *Receiver* moving around their display to observe the *Sender* from the desired view angle. Currently, our baseline achieves, 22 FPS and 172 msec latency.

4 VISITOR EXPERIENCE

For the exhibit, visitors will play the role of the *Receiver* and interact with one of our group members as the *Sender* to experience 3D teleconferencing. The visitor (*Receiver* role) can view the 3D head model of the *Sender* by using one of the 3 view-dependent displays: spherical FTVR, flat FTVR or VR headset. For the spherical FTVR display, visitors will wear active shutter glasses with a magnetic tracker as this has been shown to give the strongest realism effect [Zhou et al. 2020]. For the flat FTVR display, visitors will have a walk-up effect as we'll use a depth camera and monoscopic view-point, which, while not as good as stereoscopic, still provides a compelling sense of presence. The VR headset will give visitors an opportunity to experience what 3D teleconferencing in VR could look like at home.

Social Impact and Future Vision. The accelerated adoption of teleconferencing means this line of work will have substantial impact globally. The TeleViewDemo platform lets visitors, including researchers and the general population, experience how close this future is. We anticipate that a virtuous cycle of advances in research on encoding and decoding using 3D head reconstruction and machine learning approaches will be supported by extensions to our platform and transferred to commercial products allowing the whole world transition to 3D teleconferencing.

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